

ServiceNow Administrator Project

Team ID: SWTID-2026-9263

Date: 30 September 2026

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Problem Statement:

The current IT procurement process lacks efficiency and automation, resulting in delays and manual overhead, particularly in handling standard laptop orders. Requests for standard laptops often require configuration, but this step is prone to oversight or delay, leading to frustration among users and inefficient resource allocation within the IT department.

User Story:

As a member of the IT procurement team, I want a streamlined process for ordering standard laptops so that tasks are automatically generated and assigned to the hardware team for configuration. This will ensure that laptops are promptly configured upon arrival, reducing user wait times and minimising manual intervention.

Project Objective:

The objective of this project is to implement an automated workflow using Flow Designer to facilitate the procurement and configuration of standard laptops. By integrating automation into the process, the project aims to:

1. Create a seamless experience for users requesting standard laptops by ensuring timely configuration.
2. Reduce manual intervention and potential errors in the procurement process.
3. Improve resource utilisation within the IT department by optimising task allocation.
4. Enhance overall efficiency and productivity in IT procurement operations.

Ideation Phase
Define the Problem Statements

Date	30 SEPTEMBER 2026
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Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	2 Marks

Customer Problem Statement :

The problem statement is framed from the perspective of users requesting standard laptops and the IT procurement / hardware team. It focuses on streamlining laptop orders, automating configuration task creation and assignment, reducing manual intervention, and ensuring timely laptop configuration, as described in the project.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	an employee / standard laptop requester	place a standard laptop request and receive a configured laptop promptly	the procurement and configuration steps may involve manual intervention and delays	standard laptop requests require configuration and tasks may not be created and assigned promptly, increasing user wait time and IT workload	frustrated by waiting for a laptop to be configured and uncertain about when it will be ready
PS-2	an IT procurement / hardware team member	process approved standard laptop requests efficiently and ensure configuration work is assigned to the right team	manual task creation and assignment can cause oversight, delay and inefficient resource allocation	the configuration task needs to be generated automatically and assigned to the Hardware team after approval	concerned about avoidable manual work, delayed configuration and inefficient use of IT resources

Selected Problem Focus :

Streamline standard laptop procurement by automatically creating and assigning configuration tasks to the Hardware team after approval, reducing manual effort, delays and potential errors while ensuring laptops are configured promptly.

Ideation Phase

Empathize & Discover

Date	30 September 2026
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Maximum Marks	4 Marks

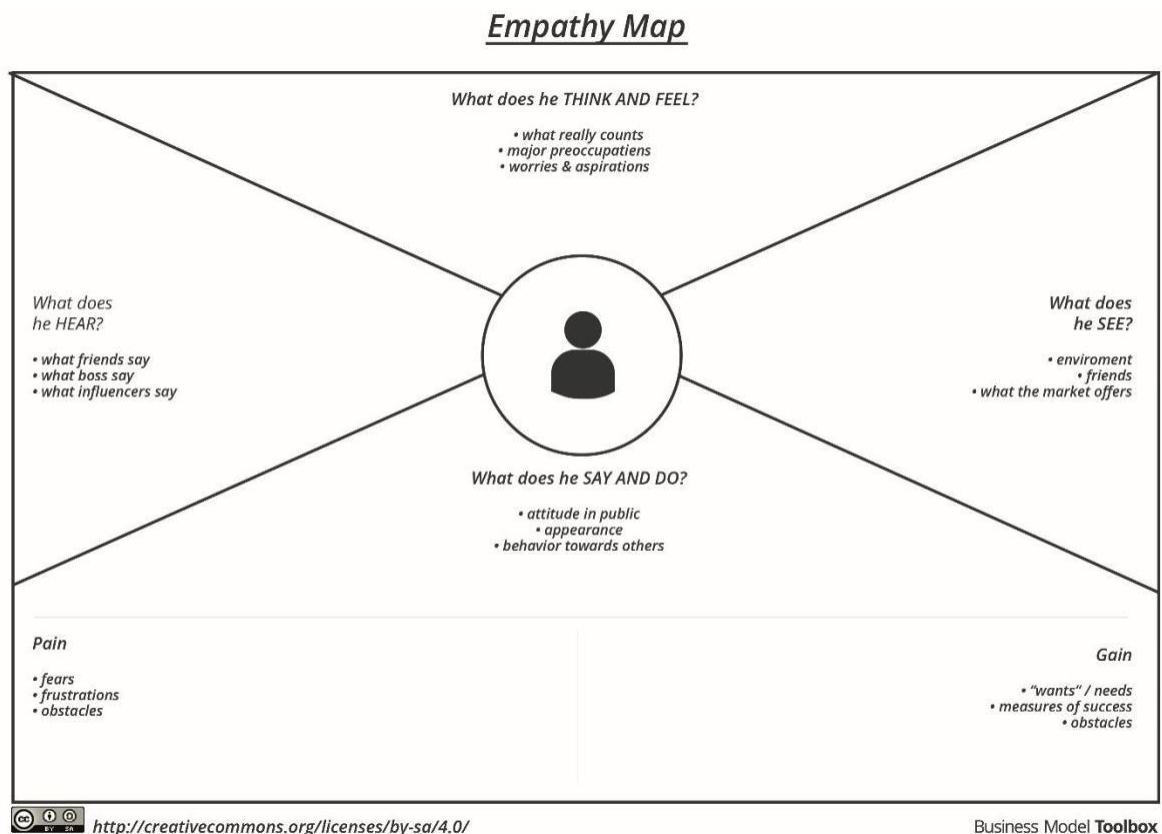
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

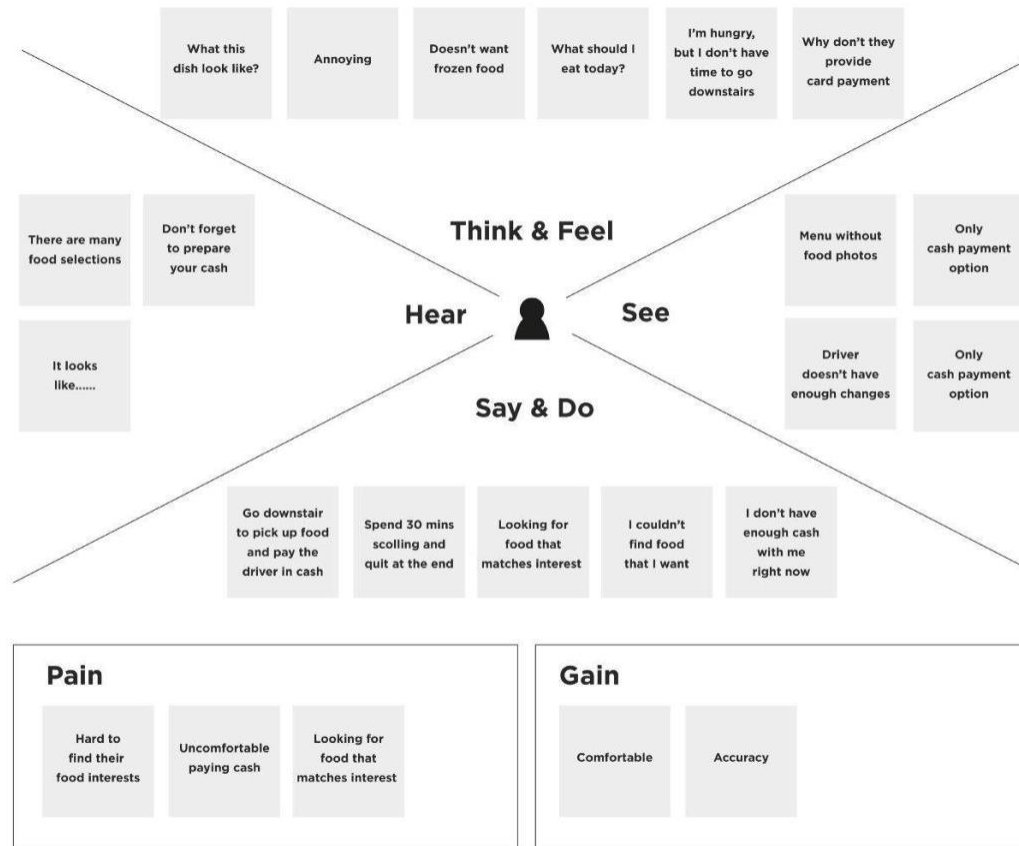
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



Reference: <https://www.mural.co/templates/empathy-map-canvas>

Example: Food Ordering & Delivery Application



Empathy Map – Standard Laptop Procurement User

SAYS • “I need a standard laptop for my work.” • “I want my laptop to be configured before I start using it.” • “I need the request and approval process to be quick.”	THINKS • Will my request be approved without delays? • Will the laptop be ready when it arrives? • I should not have to repeatedly follow up with IT.
DOES • Places a Standard Laptop request through Service Catalog. • Waits for approval and procurement processing. • Checks the request/order status and requested item record.	FEELS • Frustrated by delays and manual follow-ups. • Concerned when configuration is overlooked. • Relieved when the laptop is configured promptly.
PAINS • Manual intervention can slow the procurement process. • Configuration tasks may be missed or delayed. • Longer user wait times. • Inefficient allocation of IT resources.	GAINS • Faster laptop configuration after arrival. • Less manual intervention and fewer errors. • Clearer task assignment to the Hardware team. • A smoother and more efficient procurement experience.

Project context: The empathy map focuses on the user requesting a standard laptop, whose experience is affected by approval, procurement, configuration, and task-assignment delays.

Ideation Phase

Brainstorm & Idea Prioritization Template

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Problem Statement

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Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team selected the problem of delays and manual overhead in the standard laptop procurement and configuration process. The intended user story is to streamline ordering so that tasks are automatically generated and assigned to the Hardware team for configuration.

Step-2: Brainstorm, Idea Listing and Grouping

Idea	Proposed Solution	Impact	Feasibility	Group
Automated Catalog Task Creation	Automatically create a catalog task after the laptop request is approved and assign it to the Hardware group.	High	High	Workflow Automation
Approval-Based Workflow Trigger	Use the Service Catalog approval status to trigger the laptop configuration workflow.	High	High	Workflow Automation
Automatic Hardware Assignment	Set the Assignment group to Hardware so configuration work reaches the responsible team without manual routing.	High	High	Task Assignment & Standardisation
Standard Laptop Service Catalog	Provide a Standard Laptop catalog item so users can place a structured request through Service Catalog.	High	High	User Request & Tracking
Automatic Task Description	Populate a consistent short description and description such as 'Laptop need to	Medium	High	Task Assignment & Standardisation

	Configured' for clear task handling.			
Request Status Visibility	Use the request and Requested Item records to track approval and downstream catalog-task status.	Medium	High	User Request & Tracking
Reduce Manual Procurement Steps	Replace repeated manual task creation and assignment with a Flow Designer workflow.	High	High	Workflow Automation
Improve Resource Utilisation	Automatically route configuration work to the Hardware team to support better task allocation.	Medium	High	Operational Efficiency
Error Reduction Through Standardisation	Use fixed workflow values and a standard catalog process to reduce omissions and inconsistent routing.	Medium	High	Task Assignment & Standardisation
Post-Arrival Configuration Readiness	Create the configuration task early so the laptop can be configured promptly upon arrival.	High	High	Operational Efficiency

Idea Grouping

Group	Ideas
Workflow Automation	Automated Catalog Task Creation; Approval-Based Workflow Trigger; Reduce Manual Procurement Steps
Task Assignment & Standardisation	Automatic Hardware Assignment; Automatic Task Description; Error Reduction Through Standardisation
User Request & Tracking	Standard Laptop Service Catalog; Request Status Visibility
Operational Efficiency	Improve Resource Utilisation; Post-Arrival Configuration Readiness

Step-3: Idea Prioritization

The ideas were prioritized based on their direct connection to the stated project objective, their ability to reduce manual intervention, and their role in creating a seamless standard laptop procurement and configuration workflow.

Priority	Idea	Reason for Priority
1	Automated Catalog Task Creation	Directly addresses the core requirement to create a task after approval and route it to Hardware.
2	Approval-Based Workflow Trigger	Ensures automation follows the approved-request condition described in the project workflow.

3	Automatic Hardware Assignment	Ensures the configuration task is assigned to the Hardware team without manual intervention.
4	Standard Laptop Service Catalog	Provides the structured entry point for placing the standard laptop request.
5	Request Status Visibility	Supports tracking from request and approval through the resulting catalog task.

Selected Concept

Implement an automated ServiceNow Flow Designer workflow for Standard Laptop requests. After the service request is approved, the flow creates a Catalog Task, sets the short description and description to indicate that the laptop needs configuration, and assigns the task to the Hardware group. The Standard Laptop service catalog item is linked to this flow so that requests can progress through a consistent automated process.

Expected outcome: timely configuration, reduced manual intervention and potential errors, improved task allocation, and better overall efficiency and user experience in IT procurement.

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

CUSTOMER JOURNEY MAP

SCENARIO					USER / CUSTOMER						PROCESS OUTCOME				
Requesting and receiving a standard laptop through an automated IT procurement workflow					Employee / End User						Laptop is configured by the Hardware Team and the request is completed with minimal manual intervention				
ENTICE How does someone initially become aware of this process?	ENTER What does the person experience as they begin the process?	ENGAGE In the core moments in the process, what happens?	EXIT What does the person typically experience as the process finishes?	EXTEND What happens after the experience is over?											
Steps															
What does the person (or group) typically experience?	1. Laptop need identified	2. Employee is directed to ServiceNow	3. Open Service Catalog	4. Select Standard Laptop	5. Enter request details	6. Submit request	7. Flow Designer triggers	8. Request is validated	9. Catalog Task is created	10. Task assigned to Hardware Team	11. Laptop arrives / configuration begins	12. Hardware team configures laptop	13. Task / request status updated	14. User receives notification and laptop	15. Request completed / future requests
Interactions What interactions do they have at each step along the way? People • Places • Things / digital touchpoints	Employee identifies a need for a standard laptop for work. Touchpoint: IT policy / procurement guidance; employee's workplace	Employee receives the normal IT procurement/request guidance. Touchpoint: IT support guidance; ServiceNow access	Employee opens the ServiceNow Service Catalog. Touchpoint: ServiceNow Service Catalog / Service Portal	Employee chooses the standard laptop catalog item. Touchpoint: ServiceNow catalog item	Employee enters required details such as justification and delivery information. Touchpoint: Request form and required fields	Employee reviews the details and submits the request. Touchpoint: ServiceNow request / requested item record	ServiceNow Flow Designer detects the qualifying request and starts automation. Touchpoint: Flow Designer automation	Flow conditions check the request information and eligibility. Touchpoint: Flow Designer conditions and actions	Flow Designer automatically creates the required Catalog Task (sc_task). Touchpoint: ServiceNow Catalog Task (sc_task)	Task is routed to the Hardware Team through the appropriate Assignment Group. Touchpoint: Hardware Team Assignment Group / task queue	Hardware Team receives/locates the laptop and prepares it for configuration. Touchpoint: Physical laptop + ServiceNow task	Hardware Team configures the laptop according to the standard specifications. Touchpoint: Laptop configuration tools + task record	Task and request records are updated to reflect progress/completion. Touchpoint: ServiceNow task/request status	Employee receives the completion notification and obtains the configured laptop. Touchpoint: ServiceNow notification + configured laptop	The completed request remains traceable for future reference and similar requests. Touchpoint: ServiceNow request history / records
Goals & motivations At each step, what is a person's primary goal or motivation? “Help me...” / “Help me avoid...”	Help me understand how to request the laptop.	Help me reach the correct IT request process.	Help me find the correct standard laptop option.	Help me select a standard device confidently.	Help me provide all required information correctly.	Help me submit the request without repeated manual follow-up.	Help me know the request is being processed automatically.	Help me avoid rejection or unnecessary delay.	Help me avoid waiting for someone to remember to create the configuration task.	Help me know the right team is responsible for configuration.	Help me know the device is progressing toward readiness.	Help me receive a correctly configured laptop on time.	Help me see that the work is recorded and progressing to completion.	Help me receive the configured laptop and clear confirmation.	Help me have a traceable, repeatable process for future requests.
Positive moments What steps are enjoyable, productive, motivating, delightful, or reassuring?	Clear need and defined standard device.	Existing IT guidance reduces uncertainty.	Central catalog provides one entry point.	Standard item simplifies selection.	Form provides a structured way to give details.	Immediate submission confirmation.	Automation starts without manual hand-off.	Rules can prevent incomplete/invalid processing.	Configuration task is generated automatically.	Responsibility is clearly routed to Hardware Team.	Visible task provides operational clarity.	Laptop is configured against standard requirements.	Status is recorded in one system.	User gets a clear completion signal.	History can support future procurement.
Negative moments What steps are frustrating, confusing, costly, or time-consuming?	Employee may not know which laptop/request type to choose.	Access or unclear instructions can delay the start.	Too many catalog choices can create confusion.	Incorrect item selection can cause rework.	Missing details can cause delays or clarification requests.	Users may otherwise expect manual confirmation.	Incorrect trigger conditions could prevent automation.	Incomplete data can stop or delay fulfilment.	Without automation, task creation may be forgotten or delayed.	Wrong assignment can create a hand-off delay.	Laptop arrival may be outside the employee's control.	Manual configuration or unclear standards can delay readiness.	Status may lag behind the real work if updates are missed.	User frustration occurs if notification or delivery is delayed.	Poor records make repeat requests harder to manage.
Areas of opportunity How might we make each step better? What ideas do we have?	Provide simple guidance for choosing a standard laptop request.	Make the ServiceNow request path easy to discover.	Keep the catalog focused and clearly labelled.	Use a clear standard-laptop description and eligibility criteria.	Mark mandatory fields and validate inputs.	Show confirmation and expected next steps after submission.	Use a clear Flow Designer trigger for standard laptop requests.	Add conditions that detect missing or invalid request data.	Automatically create the Catalog Task as part of the flow.	Use a dedicated Hardware Team Assignment Group.	Track hardware arrival/configuration status in the task.	Provide a standard configuration checklist for the Hardware Team.	Automatically update task/request status at key milestones.	Send automated notifications when configuration is complete.	Use request history/metrics to identify recurring delays and improve the process.
Stage Journey phase represented by each step.	Entice	Entice	Enter	Enter	Enter	Enter	Engage	Engage	Engage	Engage	Engage	Engage	Exit	Exit	Extend

Project scope: Standard laptop procurement and configuration using ServiceNow Service Catalog and Flow Designer. This journey map is derived from the project problem statement, user story and objectives.

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

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Maximum Marks:	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Standard Laptop Request	Submit a standard laptop request with the required user, department and request details.
FR-2	Request Validation & Routing	Validate the request and automatically route eligible standard-laptop requests through the procurement workflow.
FR-3	Automated Configuration Task	Automatically create and assign a configuration task to the hardware team when the laptop procurement request reaches the required stage.
FR-4	Status & Notification Management	Update request/task status and notify relevant users or teams so configuration can be completed promptly after laptop arrival.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The workflow should provide a simple and clear process for submitting and tracking standard laptop requests with minimal manual effort.
NFR-2	Security	Only authorised users and teams should be able to submit, view, update or process procurement and configuration tasks.
NFR-3	Reliability	The workflow should consistently create and assign configuration tasks without losing request information or requiring repeated manual entry.
NFR-4	Performance	Automation should trigger workflow actions and task assignment promptly to reduce delays in laptop configuration.
NFR-5	Availability	The procurement workflow should remain available to authorised users and IT teams during normal operational hours.
NFR-6	Scalability	The workflow should support an increasing number of standard laptop requests and configuration tasks without requiring major process changes.

Project Design Phase-II

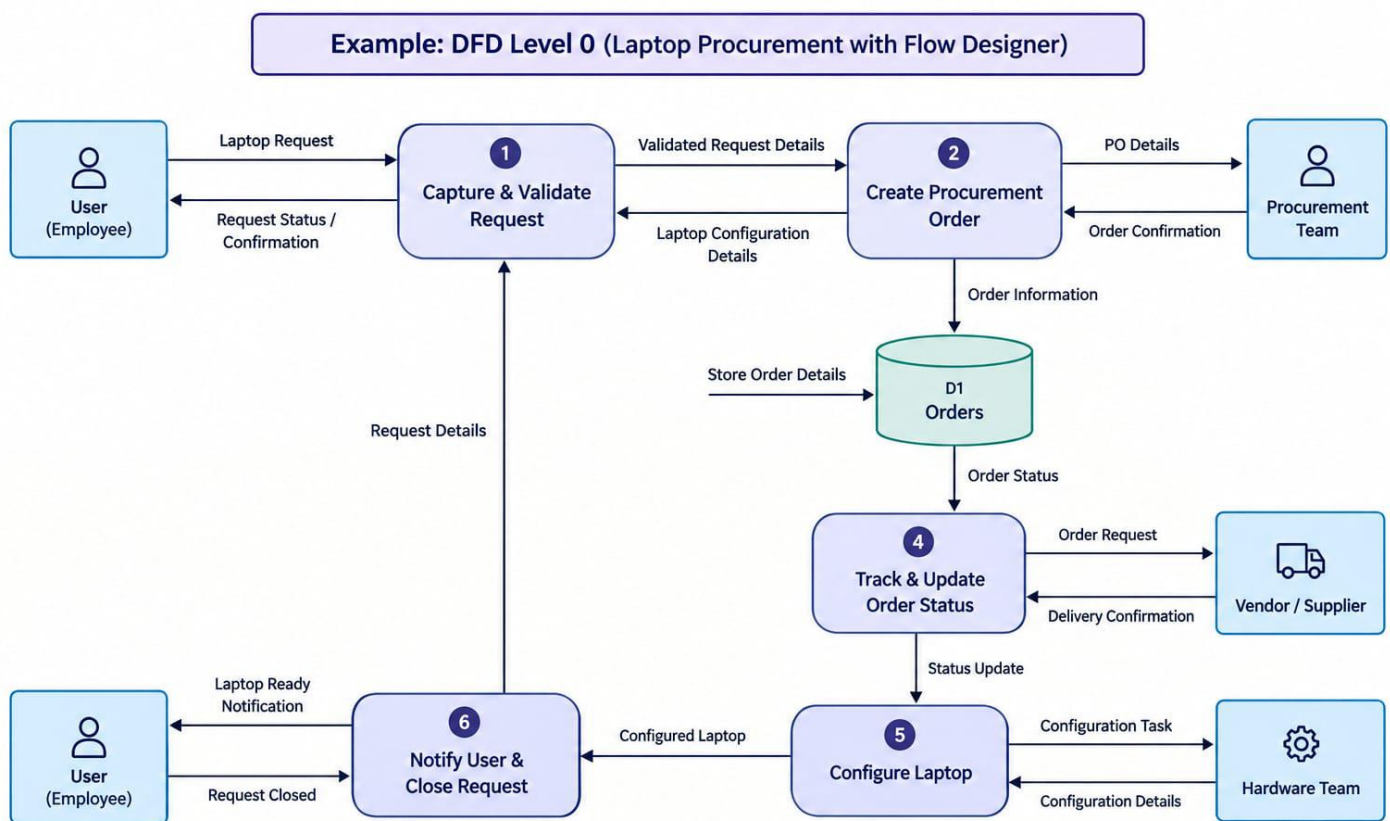
Data Flow Diagram & User Stories

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Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)



DFD flow summary:

1. The requestor submits a standard laptop request with the required details.
2. Flow Designer validates the request and initiates the automated procurement workflow.
3. The workflow creates/updates procurement and configuration tasks and assigns the configuration task to the hardware team.
4. The hardware team configures the laptop when it arrives and updates the task status.
5. The workflow updates the request status and closes the task after successful configuration.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee / Requestor	Laptop Request	USN-1	As a requestor, I can submit a request for a standard laptop with the required request details.	The request is submitted successfully and a request record is created.	High	Sprint-1
	Request Validation	USN-2	As a requestor, I can provide the required information for a standard laptop request so that the procurement flow can process it.	Required details are available and the request can proceed to the workflow.	High	Sprint-1
IT Procurement Team	Automated Procurement	USN-3	As a procurement team member, I want standard laptop requests to trigger an automated Flow Designer workflow.	A valid standard laptop request automatically starts the defined flow without manual initiation.	High	Sprint-1
	Task Generation	USN-4	As a procurement team member, I want the workflow to automatically generate the required configuration task for a standard laptop.	A configuration task is created automatically with the relevant request information.	High	Sprint-1
Hardware Team	Configuration Assignment	USN-5	As a hardware team member, I can receive an automatically assigned configuration task for a standard laptop.	The task is assigned to the hardware team and contains the information needed for configuration.	High	Sprint-1
	Laptop Configuration	USN-6	As a hardware team member, I can update the task after configuring the standard laptop.	The configuration status is updated and the completed task is recorded.	High	Sprint-2
IT Procurement Team	Status Tracking	USN-7	As a procurement team member, I can track the status of the laptop request and configuration task.	The current request/task status is available for tracking.	Medium	Sprint-2
Employee / Requestor	Request Status	USN-8	As a requestor, I can receive the updated status of my standard laptop request.	The request status reflects the latest workflow or configuration update.	Medium	Sprint-2
IT Administrator	Workflow Management	USN-9	As an administrator, I can manage and maintain the Flow Designer workflow used for standard laptop procurement.	The workflow can be maintained and changes can be applied without disrupting existing request records.	Medium	Sprint-2
IT Procurement Team	Completion & Closure	USN-10	As a procurement team member, I want completed configuration tasks to be reflected in the request so that the process can be closed without unnecessary manual follow-up.	The request reflects task completion and can be closed after successful configuration.	High	Sprint-2

Project Design Phase-II

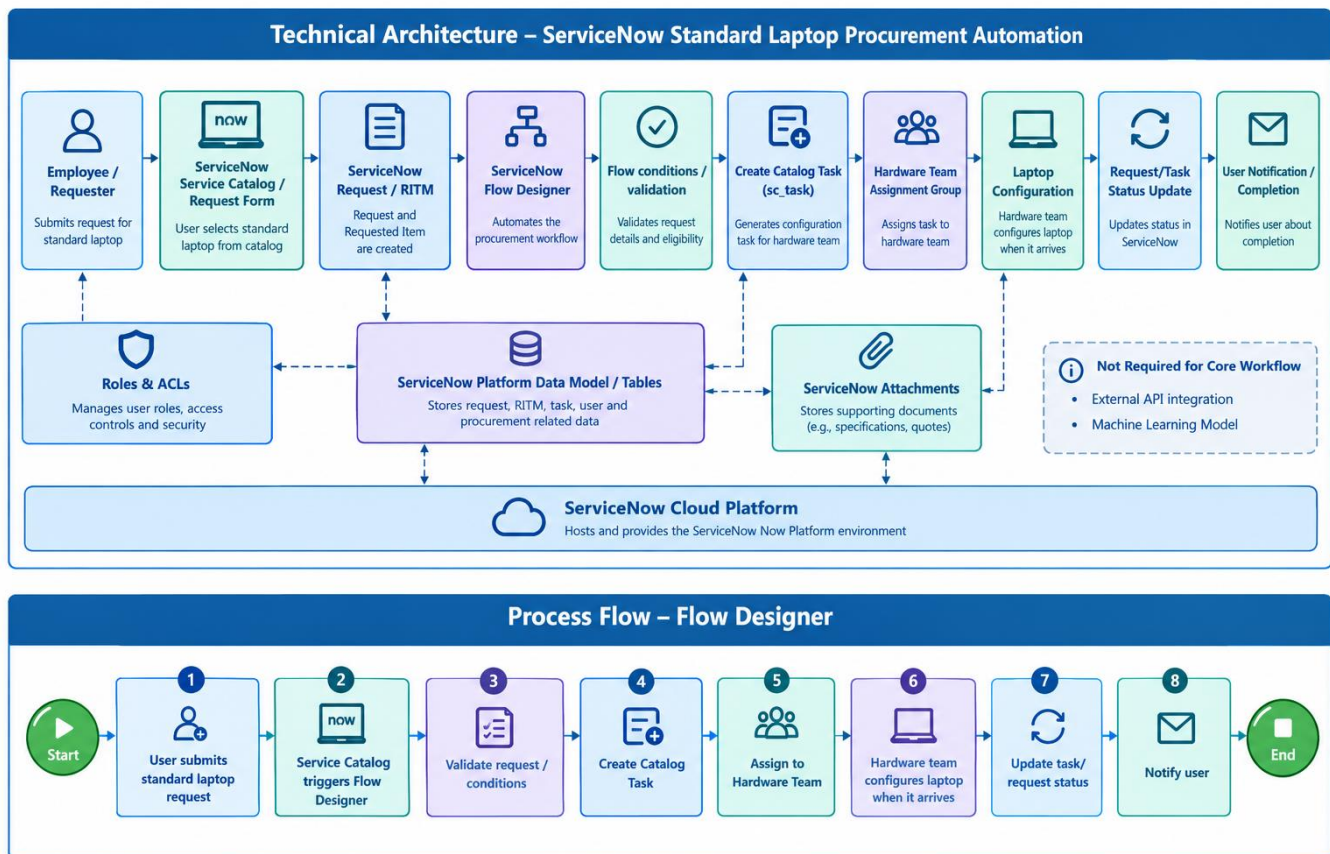
Technology Stack (Architecture & Stack)

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Technical Architecture:

The proposed solution is implemented on the ServiceNow Platform. Employees submit a standard laptop request through the ServiceNow Service Catalog. Flow Designer is triggered by the catalog request and applies the required conditions and validation before creating a Catalog Task (sc_task). The task is assigned to the Hardware Team assignment group for laptop configuration. After configuration, the task/request status is updated and the user is notified. ServiceNow platform tables/data, attachments, roles and access controls support the workflow.

Architectural Diagram – ServiceNow Standard Laptop Procurement Automation



Core flow: Employee → Service Catalog → Request/RITM → Flow Designer → Validation → Catalog Task → Hardware Team → Laptop Configuration → Status Update → User Notification.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface through which employees submit the standard laptop request and enter the required request information.	ServiceNow Service Catalog / Service Portal
2	Application Logic-1	Receives the submitted catalog request and starts the automated fulfillment process.	ServiceNow Flow Designer
3	Application Logic-2	Applies conditions/validation and creates the hardware configuration task required to fulfill the request.	Flow Designer Conditions + Actions
4	Application Logic-3	Assigns the generated catalog task to the appropriate hardware team and controls task/status progression.	Flow Designer + ServiceNow Task Management / Assignment Groups
5	Database	Stores users, requests, requested items, catalog tasks, assignments and workflow-related records.	ServiceNow Platform Data Model / ServiceNow Tables (e.g., Request, Requested Item and Task records)
6	Cloud Database	Provides the cloud-hosted data persistence used by the ServiceNow instance; direct database administration is not required by the application.	ServiceNow Now Platform Cloud Data Layer
7	File Storage	Stores supporting attachments such as laptop specifications, quotations or other procurement documents when required.	ServiceNow Attachment Storage
8	External API-1	No external API is required for the core standard laptop procurement workflow.	Not Applicable
9	External API-2	No second external API is required for the core workflow. Future integrations can be added if required.	Not Applicable / Optional REST Integration
10	Machine Learning Model	The proposed workflow is rule-based and does not require a machine learning model.	Not Applicable
11	Infrastructure (Server / Cloud)	The workflow is hosted and executed within the ServiceNow cloud environment; no separate application server is required for the core solution.	ServiceNow Now Platform – Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the core solution. The workflow is configured using ServiceNow platform capabilities and Flow Designer.	ServiceNow Flow Designer / ServiceNow Now Platform

S.No	Characteristics	Description	Technology
2	Security	Access to requests, tasks, attachments and application data is controlled using roles and Access Control Lists (ACLs). HTTPS/TLS protects data in transit, while ServiceNow provides platform-level encryption and security controls.	ServiceNow RBAC, ACLs, HTTPS/TLS, Platform Encryption
3	Scalable Architecture	The solution uses a cloud platform and reusable workflow automation. The same Flow Designer process can handle multiple standard laptop requests without creating a separate application instance for each request.	ServiceNow Now Platform + Flow Designer
4	Availability	The application runs on the ServiceNow cloud platform rather than on a project-managed local server. ServiceNow provides cloud-hosted platform infrastructure and high-availability capabilities.	ServiceNow Now Platform Cloud
5	Performance	Automation reduces manual hand-offs by creating and assigning the hardware task immediately after the qualifying request is submitted. This reduces processing delay and improves fulfilment throughput.	Flow Designer Automation + ServiceNow Task Management

References:

ServiceNow – Now Platform Reference Guide: <https://www.servicenow.com/content/dam/servicenow-assets/public/en-us/doc-type/resource-center/ebook/ebk-now-platform-reference-guide.pdf>

ServiceNow – Securing the Now Platform: <https://www.servicenow.com/content/dam/servicenow-assets/public/en-us/doc-type/resource-center/ebook/ebk-how-servicenow-delivers-safe-and-secure-cloud-services.pdf>

ServiceNow Developers – Now Platform / Data Model: <https://developer.servicenow.com/>

ServiceNow – C4 Model reference used by the template: <https://c4model.com/>

IBM Cloud Architecture reference retained from the template: <https://www.ibm.com/cloud/architecture>

AWS Architecture reference retained from the template: <https://aws.amazon.com/architecture>

Project Design Phase Problem – Solution Fit

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Problem – Solution Fit:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Primary Target: IT Procurement & Hardware Configuration Group. End Users: Enterprise Employees requesting Standard Laptops. Stakeholders: IT Service Desk Managers.	2. JOBS-TO-BE-DONE / PROBLEMS J&P Core Job: Rapidly configure and fulfill standard laptop requests upon approval. Key Problem: Lack of automated task generation leads to configuration oversights, missing task assignments, and frustrated end users	3. TRIGGERS TR Employee orders 'Standard Laptop' in Service Catalog. Manager approves the Request (e.g., REQ0010001). Catalog Request moves to 'Approved' state.	Explore AS, differentiate	
	5. AVAILABLE SOLUTIONS AS High volume of incoming tickets leading to manual routing crassing tickets. Manual technician routing causing unassigned tasks. Slow approval handoffs stalling.	6. CUSTOMER CONSTRAINTS CC High volume of incoming tickets leading to manual review delays. Manual technician routing causing unassigned tasks. Slow approval handoffs stalling dispatch.	9. PROBLEM ROOT CAUSE BC Manual Workflows: Request processing via manual emails and unassigned tickets. Static execution scripts lacking dynamic flow triggers.		7. BEHAVIOUR BE Hardware technicians manually inspect and pick up queues periodically. Requesters repeatedly email IT desk to track laptop configuration. 8. CHANNELS OF BEHAVIOUR CH 8.1 ONLINE Online: ServiceNow Service Portal, Flow Designer, Agent Workspace. 8.2 OFFLINE Offline: Physical IT Staging Desk & Asset Inventory Room
Focus on J&P, tap into BE, understand RC					Focus on J&P, tap into BE, understand RC
Identify strong TR & EM	9. PROBLEM ROOT CAUSE RC Service catalog items were not linked to modern automated workflow triggers. Approval actions failed to trigger automatic creation and assignment of SCTASK.	10. YOUR SOLUTION SL Standard Laptop Task Flow: Flow Designer workflow triggered by Catalog Item. Automatically generates Catalog Task (SCTASK) assigned to Hardware group. Auto-populates description: 'Laptop needs to Configured'.			Extract online CH of BE

Project Design Phase
Proposed Solution

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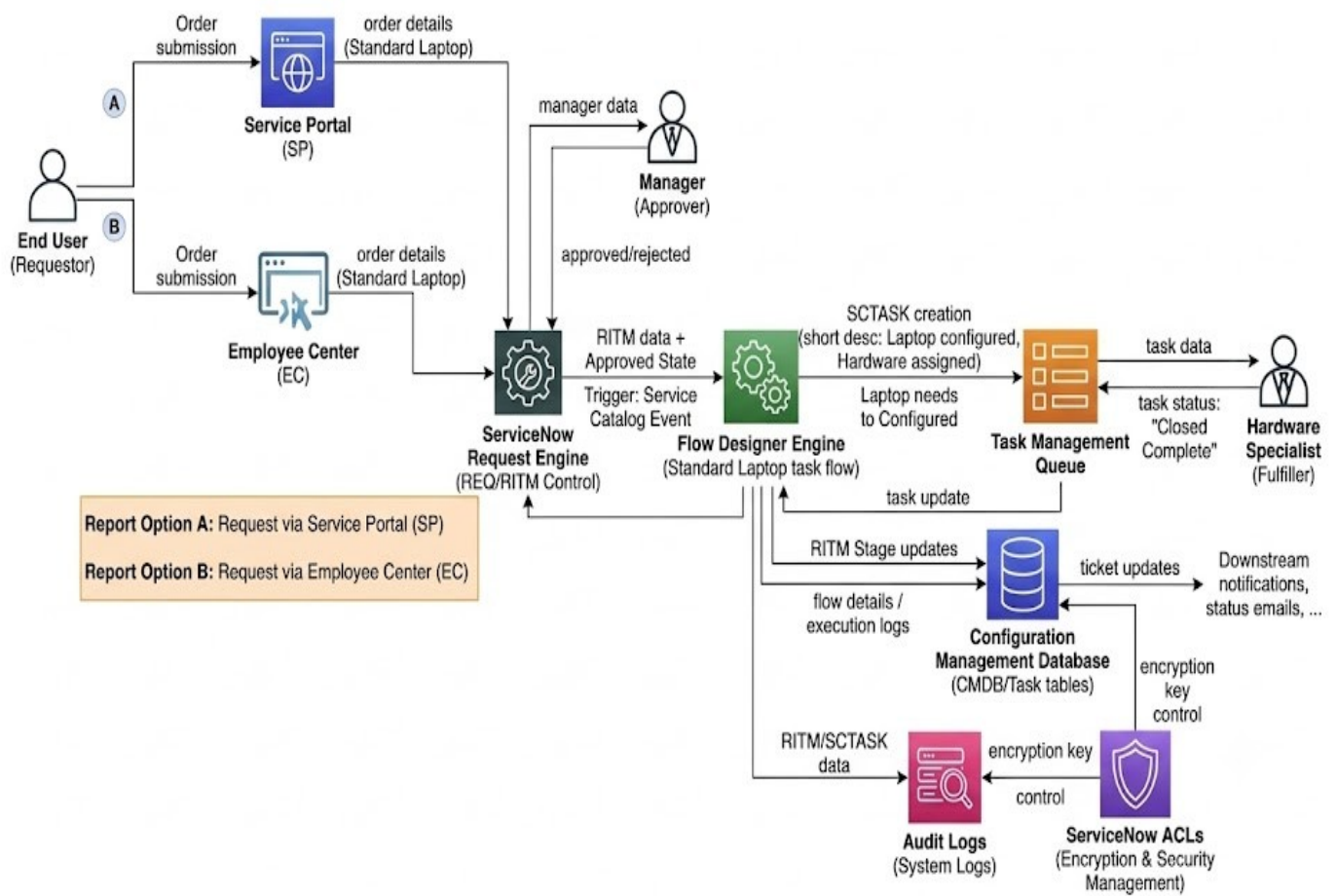
Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	In traditional IT Service Management environments, purchasing and provisioning standard employee hardware (such as laptops) relies heavily on manual ticket creation, human handoffs, and manual queue polling. This results in missing task assignments, extended fulfillment lead times, operational bottlenecks, and lack of real-time visibility for both requesters and IT technicians.
2.	Idea / Solution description	Implementation of an automated workflow engine using ServiceNow Flow Designer for the <i>Standard Laptop</i> Catalog Item. Upon manager approval of a Service Catalog Request, the system automatically triggers a dedicated flow that creates a Catalog Task (SCTASK), pre-populates configuration parameters (e.g., " <i>Laptop needs to Configured</i> "), and instantly routes the task to the designated Hardware Configuration Group without manual intervention.
3.	Novelty / Uniqueness	●Zero-Touch Dispatch: Completely eliminates manual queue assignment by linking Service Catalog approvals directly to task generation. ●No-Code Workflow Logic: Built natively using ServiceNow Flow Designer, ensuring high maintainability and easy customization without legacy script overhead. ●Event-Driven Execution: Instant state-triggered routing ensures end-to-end task tracking from order placement to hardware staging.
4.	Social Impact / Customer Satisfaction	●Reduced Employee Downtime: Accelerates new-hire onboarding and hardware replacement cycles by reducing procurement turnaround times. ●Enhanced Transparency: Requesters receive clear, real-time updates regarding order approval and configuration progress. ●Minimized Human Error: Prevents overlooked orders and lost configuration tasks, boosting trust in internal IT support services.
5.	Business Model (Revenue Model)	●Internal Cost Reduction (Operational Efficiency): Saves technician working hours by automating routine administration tasks, yielding direct operational cost savings. ●Enterprise ITSM SaaS Integration: Reusable module framework that can be monetized as an IT Operations Management (ITOM) / ITSM workflow package for enterprise client instances.
6.	Scalability of the Solution	●Multi-Hardware Support: Flow logic can be effortlessly cloned and extended to handle mobile devices, monitors, and specialized peripheral requests. ●Cross-Departmental Expansion: The same workflow architecture scales across HR Onboarding, Access Management, and Facilities asset provisioning. ●Enterprise Ready: Fully scalable to process hundreds of concurrent request streams across global enterprise divisions within ServiceNow.

Project Design Phase Solution Architecture

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Solution Architecture:



Project Planning Phase

Project Planning

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Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Flow Creation	USN-1	As an IT procurement team member, I can create the Standard Laptop Task flow in Flow Designer.	2	High	_____
Sprint-1	Flow Creation	USN-2	As the system, I can trigger the flow from the Service Catalog request.	2	High	_____
Sprint-1	Flow Creation	USN-3	As the system, I can create a Catalog Task from the Requested Item record.	3	High	_____
Sprint-1	Flow Configuration	USN-4	As the system, I can update the task short description and description to indicate laptop configuration is required.	2	High	_____
Sprint-1	Flow Configuration	USN-5	As the system, I can assign the task to the Hardware group when the request is approved.	3	High	_____
Sprint-2	Service Catalog Integration	USN-6	As an administrator, I can assign the created flow to the Standard Laptop service catalog item.	3	High	_____
Sprint-2	Service Catalog	USN-7	As a user, I can place a Standard Laptop order through the Service Catalog.	2	High	_____
Sprint-2	Approval	USN-8	As an approver, I can approve the requested laptop item.	2	High	_____
Sprint-2	Verification	USN-9	As the system, I can generate and update the Catalog Task with the required assignment information.	3	High	_____
Sprint-2	Verification	USN-10	As an IT team member, I can verify the task status, short description	2	Medium	_____

			and assigned Hardware group.			
--	--	--	---------------------------------	--	--	--

Planned total story points = 24.

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	12	6 Days	05 Oct 2026	10 Oct 2026	12	10 Oct 2026
Sprint-2	12	6 Days	12 Oct 2026	17 Oct 2026	12	17 Oct 2026

Velocity:

Total story points = 24

No. of Sprints = 2

Velocity = $24 / 2 = 12$ Story Points per Sprint

Burndown Chart:

A burndown chart is a graphical representation of work left to do versus time. For this project, the planned remaining story points are 24 at project start, 12 after Sprint-1, and 0 after Sprint-2.

Project Workflow Reference

Create the “Standard Laptop Task” Flow in Flow Designer.

Use the Service Catalog trigger.

Create a Catalog Task using the Requested Item record.

Set the task description to “Laptop need to Configured”.

Assign the task to the Hardware group and set Approval as Approved.

Attach the flow to the Standard Laptop service catalog item.

Place the Standard Laptop order through Service Catalog.

Approve the request and verify the generated Catalog Task.

PROJECT PLANNING LOGIC

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Date: 30 September 2026

A Sprint is a fixed period or duration in which a team works to complete a set of tasks.

An Epic is a big task or project that is too large to complete in one sprint. It is broken down into smaller tasks (stories) that can be completed over multiple sprints.

A Story is a small task. It is part of an Epic.

A Story Point is a number that represents how much effort a story takes to complete (usually in the form of the Fibonacci series).

1 - Very Easy task

2 - Normal task

3 - Moderate task

5 - Difficult task

Story point values used: 1, 2, 3, 5

Sprint 1

Flow Creation and Configuration (Epic 1)

Create Flow (USN1) 2

Configure Service Catalog trigger (USN2) 2

Create Catalog Task (USN3) 3

Configure task description (USN4) 2

Assign task to Hardware group and approval condition (USN5) 3

Total Story Point in Sprint 1 = $2 + 2 + 3 + 2 + 3 = 12$

Sprint 2

Service Catalog Integration (Epic 2)

Assign Flow to Standard Laptop catalog item (USN6) 3

Place Standard Laptop order (USN7) 2

Approve requested laptop item (USN8) 2

Verify generated Catalog Task and Hardware assignment (USN9) 3

Verify task status, short description and assignment group (USN10) 2

Total Story Point in Sprint 2 = $3 + 2 + 2 + 3 + 2 = 12$

Total Story Points

Sprint 1 = 12

Sprint 2 = 12

Total story points = $12 + 12 = 24$

No. of Sprints = 2

Velocity = Total Story Points Completed / Number of Sprints

Velocity = $24 / 2$

12 Story Points per Sprint

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	31 January 2025
Team ID	SWTID-2026-9263
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Team members identified the manual effort involved in requesting, reviewing, approving, and fulfilling standard laptop orders.

Problem Statement: Standard laptop procurement requests are handled through multiple manual steps, which can cause delays, inconsistent information, repeated follow-ups, and limited visibility into request status.

Goal: Streamline the procurement process by using ServiceNow Flow Designer to automate standard laptop requests from submission through approval and fulfillment.

Step-2: Brainstorm, Idea Listing and Grouping

1	Create a standard laptop request form with predefined laptop options and required requester details.
2	Automatically validate mandatory information before the request enters the workflow.
3	Route the request to the appropriate approver based on predefined business rules.
4	Send automated notifications when a request is submitted, approved, rejected, or completed.
5	Create fulfillment tasks automatically for the IT/procurement team after approval.
6	Maintain a central request record so users and support teams can track progress.
7	Use Flow Designer to connect approvals, task creation, notifications, and status updates in one workflow.

Step-3: Idea Prioritization

High Priority: Standardized request form, automated approval routing, automatic fulfillment task creation, and status notifications.

Medium Priority: Request tracking dashboard and additional notification templates.

Future Enhancement: Reporting and analytics for procurement turnaround time, request volumes, and recurring laptop demand.

Prioritization Basis: Ideas were grouped according to their direct impact on reducing manual work, improving process consistency, and increasing visibility.

Acceptance Testing

UAT Execution & Report Submission

Date	03 October 2026
Team ID	SWTID-2026-9263
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the **Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer** project at the time of the release to User Acceptance Testing (UAT). It covers how the automated laptop request flow (request submission, manager approval, procurement task creation, vendor order and requester notification) was validated.

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved during testing of the automated laptop ordering workflow.

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	10	4	2	3	19
Duplicate	1	0	3	0	4
External	2	3	0	1	6
Fixed	11	2	4	20	37
Not Reproduced	0	0	1	0	1
Skipped	0	0	1	1	2
Won't Fix	0	5	2	1	8
Totals	24	14	13	26	77

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and untested across each area of the laptop procurement automation.

Section	Total Cases	Not Tested	Fail	Pass
Laptop Catalog Item & Request Form	7	0	0	7
Flow Designer Automation (Standard Laptop Order)	51	0	0	51
Security & Access Control	2	0	0	2
Vendor Order Fulfilment	3	0	0	3
Exception Reporting	9	0	0	9
Final Report Output	4	0	0	4
Version Control	2	0	0	2
Totals	78	0	0	78

Full Stack Development with MERN

Project Documentation

1. Introduction

• **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

• **Team Members:**

- Tarangini PR - Team Lead & ServiceNow System Administrator (Overall project orchestration, instance setup, and deployment verification)
- Charunethra T – ServiceNow Process Architect (Workflow logic design, Flow Designer trigger & task action setup)
- Prasanthi P – Service Catalog Specialist (Catalog item configuration, Process Engine linking in Maintain Items)
- Varshini M – Quality Assurance & Testing Lead (End-to-end user request testing, approval routing validation, and SCTASK assignment verification)
- Laya Abraham – Technical Documentation & UI Specialist (Architecture diagramming, technical document creation, and demo video production)

2. Project Overview

• **Purpose:** The current IT procurement process lacks automation, causing delays and manual overhead when processing standard laptop requests. This project implements an automated workflow within ServiceNow using Flow Designer to instantly create, configure, and assign catalog fulfillment tasks (SCTASK) to the Hardware team upon request approval.

Features:

- Automated trigger execution on Service Catalog submissions.
- Auto-generation of Catalog Tasks (SCTASK) with short description set to "Laptop needs to Configured".
- Direct task routing and assignment to the Hardware group.
- Automatic state transition updates linking Request (REQ), Requested Item (RITM), and Catalog Task (SCTASK).
- Seamless integration with Service Portal and Employee Center catalogs.

3. Architecture

- **Frontend:** ServiceNow Service Portal and Employee Center UI components providing the catalog interface for end users to browse hardware items and submit orders.

- **Backend:** ServiceNow Platform Process Engine and Flow Designer (Standard Laptop task flow running as System User) executing backend workflow automation.
- **Database:** ServiceNow relational relational record schema mapping parent-child relationships across the Request (sc_request), Requested Item (sc_req_item), and Catalog Task (sc_task) database tables.

4. Setup Instructions

- **Prerequisites:** ServiceNow Developer Instance (Zurich/Australia releases or above).
System Administrator (admin) role privileges.
- **Installation:**
- Log into your ServiceNow Developer Instance.
- Navigate to Process Automation --> Flow Designer.
- Create a new flow titled Standard Laptop task in the Global scope with Run As set to System User.
- Set Trigger to Service Catalog.
- Add Action Create Catalog Task, drag/drop the Requested Item Record, and set the fields:
 - Short Description: Laptop needs to Configured
 - Description: Laptop needs to Configured
 - Assignment Group: Hardware
 - Approval: Approved
- Save and Activate the flow.
- Navigate to Maintain Items, open Standard Laptop, go to the Process Engine tab, and assign the flow Standard Laptop task.

5. Folder Structure

Client (Presentation Layer):

- Service Catalog / Hardware – Catalog view and item ordering forms.
- Service Portal (SP) / Employee Center (EC) – User-facing order entry portals.

Server (Business Logic Layer):

- Flow Designer / Flows / Standard Laptop task – Main automation flow logic.
- Data Schema / Tables – sc_request, sc_req_item, sc_task records.

6. Running the Application

- Triggering the Workflow: Navigate to Service Catalog --> Hardware --> Standard Laptop and click Order Now.
- Approving Request: Open the generated Request record (REQ), scroll to the Approvers section, right-click the record, and click Approve.
- Verifying Output: Open the child Requested Item (RITM) record, scroll to Catalog Tasks, and verify the automatically generated SCTASK.

7. API Documentation

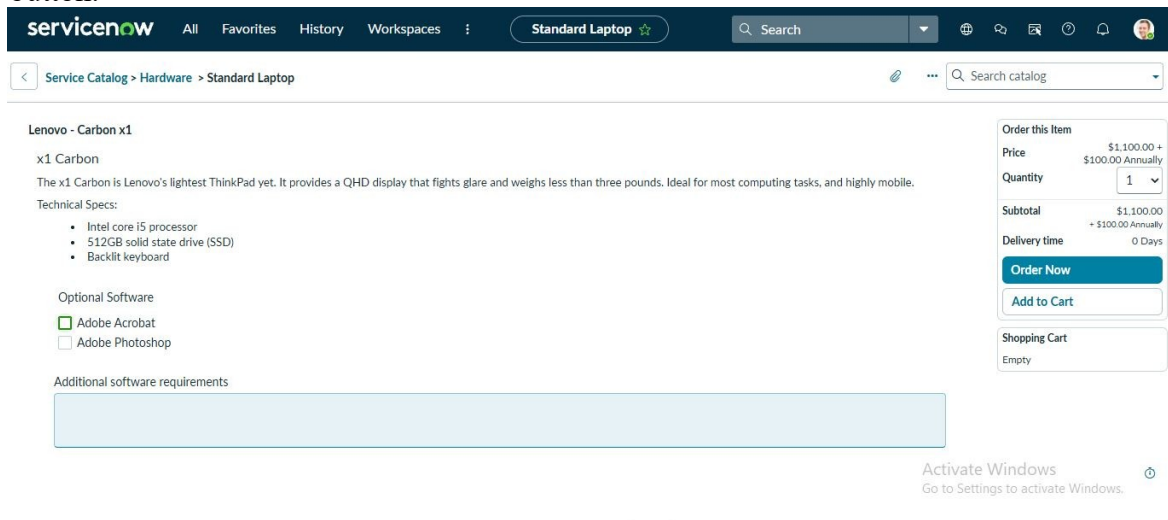
- Trigger Event: Service Catalog Submission Event
- Internal Table REST/Data APIs:
 - POST /api/now/table/sc_request – Standard Catalog Request generation.
 - POST /api/now/table/sc_req_item – Requested Item record instantiation.
 - POST /api/now/table/sc_task – Automated Catalog Task generation via Flow Designer.
- Payload Fields:
 - short_description: "Laptop needs to Configured"
 - assignment_group: "Hardware"
 - approval: "Approved"

8. Authentication

- User Authentication: Handled natively via ServiceNow Role-Based Access Control (RBAC) and system login sessions.
- Execution Authorization: The Standard Laptop task flow is configured to Run As: System User to ensure task generation executes elevated automation privileges regardless of the end user's system role.

9. User Interface

Service Catalog order submission page for Standard Laptop showing the "Order Now" button.



The screenshot shows the ServiceNow Service Catalog interface. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a search bar. The breadcrumb trail indicates the path: 'Service Catalog > Hardware > Standard Laptop'. The main content area displays the 'Lenovo - Carbon x1' product. The description states: 'The x1 Carbon is Lenovo's lightest ThinkPad yet. It provides a QHD display that fights glare and weighs less than three pounds. Ideal for most computing tasks, and highly mobile.' Technical specifications listed include: Intel core i5 processor, 512GB solid state drive (SSD), and Backlit keyboard. Under 'Optional Software', there are checkboxes for 'Adobe Acrobat' (checked) and 'Adobe Photoshop' (unchecked). A section for 'Additional software requirements' is present but empty. On the right sidebar, the 'Order this Item' section shows a price of \$1,100.00 + \$100.00 Annually, a quantity of 1, a subtotal of \$1,100.00 + \$100.00 Annually, and a delivery time of 0 Days. The 'Order Now' button is highlighted in blue. Below it is an 'Add to Cart' button. The 'Shopping Cart' section shows it is empty. At the bottom right, there is a 'Activate Windows' watermark.

Order placement confirmation screen displaying Request Number REQ0010001 and estimated delivery details.

servicenow All Favorites History Admin : Order Status: REQ0010001 Search

Order Status [Back to Catalog](#) [Continue Shopping](#) [Home](#)

Thank you, your request has been submitted

Order Placed: 2026-10-01 00:21:03
Request Number: [REQ0010001](#) ☆
Estimated Delivery Date of Complete Order: 2026-10-01

Description (Includes Annual Charges)	Delivery Date	Stage	Price (ea.)	Quantity	Total
Lenovo - Carbon x1	2026-10-01		\$1,100.00 +\$100.00 Annually	1	\$1,100.00 +\$100.00 Annually
Total					\$1,100.00 +\$100.00 Annually

[Back to Catalog](#) [Continue Shopping](#) [Home](#)

Service Request record view showing state set to Pending Approval.

servicenow All Favorites History Admin : Order Status: REQ0010001 Search

Request REQ0010001 [Follow](#) [Update](#) [Add New Item](#) [Cancel Request](#) [Copy](#) [Delete](#)

Number: REQ0010001
Requested for: System Administrator
Location:
Due date: 2026-10-01 00:21:02
Price: \$1,100.00
Description:
Short description:
Special instructions:
Work notes: Work notes

Opened: 2026-10-01 00:21:03
Opened by: System Administrator
Approval: Requested
Request state: Pending Approval

Post

Request record (REQ) showing the Approvers tab and manager approval action.

servicenow All Favorites History Admin : Request - REQ0010001 Search

Request REQ0010001 [Discuss](#) [Follow](#) [Update](#) [Add New Item](#) [Cancel Request](#) [Copy](#) [Delete](#)

Impact: 3 - Low
Opened by: System Administrator
Priority: 4 - Low
State: Open

Update Add New Item Cancel Request Copy Delete

Requested Items (1) Approvals (1) Recurring Prices (1)

State:
Approval for = REQ0010001
Show Matching
Filter Out
Copy URL to Clipboard
Copy sys_id
Assign Tag
Approve
Reject

Approver	Comments	Created
Eric Schroeder		2026-10-01 00:21:08

1 to 1 of 1

10. Testing

Strategy: End-to-end integration testing in a ServiceNow personal developer instance.

Test Cases:

- Order Submission: Validate that ordering a standard laptop generates a valid REQ and RITM record.
- Approval Trigger: Validate that approving the request triggers the Standard Laptop task flow.
- Field Population: Verify that SCTASK short description equals "Laptop needs to Configured" and assignment group equals "Hardware".

11. Screenshots or Demo

Creating a new workflow titled "Standard laptop task" in Workflow Studio.

The screenshot shows the 'New Flow' configuration screen in ServiceNow Workflow Studio. The left sidebar contains a 'Return home' button and a visual flow diagram with two steps. The main area is titled 'Let's get the details for your flow' and contains the following fields:

- Flow name:** Standard laptop task
- Application:** Global
- Description:** Describe your flow.
- Hide additional properties:** A link to expand more options.
- Protection:** -- None --
- Run as:** System user

At the bottom right, there are 'Cancel' and 'Build flow' buttons. An 'Activate Windows' watermark is visible in the bottom right corner.

Configuring the "Create Catalog Task" action with Short Description, Assignment group (Hardware), and Approval parameters.

The screenshot shows the 'Create Catalog Task' action configuration screen in ServiceNow Workflow Studio. The left sidebar contains a 'Return home' button and a visual flow diagram with one step. The main area is titled 'Standard laptop task' and contains the following fields:

- Action:** Create Catalog Task
- Table Name:** Catalog Task [sc_task]
- Requested Item (Requested Item):** Trigger -> Requested Item
- Short Description:** Laptop need to Configured
- Description:** Laptop need to Configured
- Assignment group:** Hardware
- Approval:** Approved
- Wait:** ☒

At the bottom right, there are 'Test' and 'Activate' buttons. An 'Activate Windows' watermark is visible in the bottom right corner.

Linking the active "Standard laptop task" flow to the Catalog Item under the Process Engine tab in Maintain Items.

servicenow All Favorites History Admin Catalog Item - Standard Laptop Search

Catalog Item Standard Laptop Copy Try It Update Edit in Catalog Builder Delete

Build and modify items faster with the improved [Catalog Builder](#).

Name: Standard Laptop Application: Global
Catalogs: Service Catalog Active: ☒
Category: Hardware Roles:
State: -- None -- Fulfillment automation level: Unspecified
Checked out: -- None --
Owner:
Secondary owners:

Item Details **Process Engine** Picture Pricing Portal Settings

Flow: Standard laptop task
Workflow:

Requested Item record (RITM0010001) showing the generated Catalog Task under the Catalog Tasks tab.

servicenow All Favorites History Admin Requested Item - RITM0010001 Search

Requested Item RITM0010001 Discuss Follow Update Delete

Impact: 3 - Low
Opened by: System Administrator
State: Open

Update Delete

Related Links
[Flow Context](#)

Catalog Tasks (1) Approvers

Request Item = RITM0010001

Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
SCTASK0010001	Hardware	(empty)	Laptop need to Configured	2026-10-01 00:25:00	(empty)	

1 to 1 of 1

Auto-generated Catalog Task record (SCTASK0010001) assigned to the Hardware group for fulfillment.

servicenow All Favorites History Admin : Catalog Task - SCTASK0010001 Search

Catalog Task
SCTASK0010001

Number: SCTASK0010001

Assigned to: [Search]

Configuration item: [Search]

Active: ☒

Approval: Approved

Priority: -- None --

State: Open

Request item: RITM0010001 [Search]

Requested for: System Administrator [Search]

Short description: Laptop need to Configured

Description: Laptop need to Configured

Work notes: Work notes

☐ Comments (Customer visible) **Post**

Activities: 1 System

Field changes • 2026-10-01 00:25:00

12. Known Issues

- Manual right-click approval in demo instances requires Administrator session overriding.
- Flow execution delays of 1–2 seconds may occur if background job workers are heavily loaded.

13. Future Enhancements

- Automated email notifications sent to the Hardware group upon task creation
- Integration with asset management tables to auto-reserve available laptop serial numbers.
- Adding SLA workflow timers to track hardware configuration turn-around times.

Appendix

Github link:

<https://github.com/laya-dev01/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-with-Flow-Designer.git>

Demo Link:

https://drive.google.com/file/d/1e_yWPcsgBeQFKbYL0_XYDcya4kBo3MeV/view?usp=drive_link